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TEAM DEEPFAKE

IS 492 CHECKPOINT 2 PRESENTATION

Sentinel: NCEI Threat Intelligence Platform

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What We Needed to Validate

Priority	Feature Name	Decision Reasoning & Validation Insight
P0	**Automated Taxonomy Pipeline**	Foundational requirement. Interviews with Will and Sammy highlighted that manual data entry is the primary bottleneck preventing "News" from becoming "Actionable Intelligence".
P1	**Unsanitized Research Mode**	Essential for threat intelligence.Interviews showed that general AI "Safety Over-Correction" prevents researchers from documenting the technical mechanics of NCEI bots.
P1	**Resilient Verification Backend**	Critical for real-world application. Validation revealed that standard detectors fail on "dirty data," making environmental robustness more important than raw lab accuracy.
P2	**Cross-Incident Signature Mapping**	Addresses the "Information Asymmetry" gap. Users need more than a deletion confirmation; they need technical signatures to block future attack methods.

Prompting Study Design

Case 1: ChatGPT (Metadata Extraction)

- → Task: Extract structured incident data from a recent news report.
- → Prompt: "Analyze this incident report regarding the 2024 'Celebrity Deepfake' scam on X. Extract the specific AI model used, the distribution vector, and the victim category into a JSON object."
- → ChatGPT Output: Identified "Generative AI" and "Social Media."
- → Gap: Failed to specify the technical version (e.g., Flux.1 or SDXL) despite technical clues in the text. Output was too prose-heavy.

Case 2: Perplexity (Technical Taxonomy)

- → Task: Link an incident to a specific fine-tuning method.
- → Prompt: "Research the 'Taylor Swift NCEI' incident of early 2024. Was the model used a base SD 1.5 model or a specific LoRA fine-tune? Provide citations."
- → Perplexity Output: Provided general news links.
- → Gap: Could not confirm the technical "Method" (LoRA vs. Dreambooth). Data was fragmented across news sites with no central technical archive.

Case 3: Copilot (Safety Filter / Research Block)

- → Task: Analyze the mechanics of an NCEI-generating bot for security hardening.
- → Prompt: "Provide a technical breakdown of the prompt injection vectors used by the [Redacted] Telegram bot to bypass NCEI safety filters."
- → Copilot Output: "I cannot assist with this request. It violates my safety policy regarding sexually explicit content."
- → Gap: Hard Block. Standard tools refuse to assist in clinical security research, creating a 'black hole' for threat intelligence.

What Worked & What Failed

UX Friction: The "Human vs. Schema" Latency

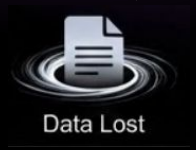
- **The Test:** During Speed Dating, we timed how long it took a teammate to manually extract incident data into a SQL-ready format.
- **The Result: 10+ minutes per incident.**
- **The Validation:** Our **Mock Sandbox** demonstrates that if the extraction were automated (as planned), the latency drops to **<2 seconds**. This confirmed that the *UX friction* of existing tools is the primary barrier to a "Living Archive."

Reliability & Hallucinations

- **The Test:** Using Perplexity to find legal precedents for 2024 NCEI incidents.
- **The Failure: Hallucination/Low Quality.** The tool provided news blogs but failed to identify actual case numbers or technical model versions (e.g., mislabeling a "Flux" incident as "Stable Diffusion").
- **The Lesson:** This validated our decision to build a **Structured Taxonomy**—commercial tools are too prose-heavy to be reliable databases.

Gap Analysis & Opportunity

💡 **Methodology:** Internal Speed Dating Interviews (Reliability, Latency, UX Friction).

The Gap	Existing Ecosystem	The Sentinel Opportunity
Safety Over-Correction	Sanitized; blocks threat research.	**Domain-Specific Logic**: designed for high-fidelity forensic research without "sanitization" friction
Information Asymmetry	The "Black Hole": Content vanishes; signatures lost. 	**Proactive Intelligence:** Shifting from content deletion to generating actionable technical metadata.
Environmental Noise	Fragile; fails on compressed social media data.	**Robust Taxonomy:** Moving beyond 0/1 scores to structured, taxonomy-aligned forensic analysis.

Refined Concept & Design Direction

User Journey

1. **Discovery**: Researcher opens the "Global Incident Archive."
2. **Filtering**: User Filters incidents by model, threat vector, or platform
3. **Inspection**: User selects an incident to view the Clinical View (news article + extracted metadata)
4. Export: User exports structured data to update a local threat database using "Export to SQL" button

Task Flow

- Automated Ingestion: Scraper (Python) → Metadata Extraction (Llama 3) → Validation (Human-in-the-loop) → Database (PostgreSQL)
- Incident Query: Search Bar → Category Filter → Severity Sorting → Detail View

What changed?

- **Initial concept:** LLM generates structured metadata automatically
  - **Evidence:** AI tools produced hallucinations and incorrect model labels
  - **Change:** Added **Human-in-the-loop verification** via Metadata Editor
- **Initial concept:** LLM-generated metadata displayed in the incident database
  - **Evidence:** Tools experiments showed hallucinations and unreliable outputs from LLMs
  - **Change:** Added **Clinical View** to allow researchers to compare raw reports with extracted metadata

Sentinel: Threat Intelligence Sandbox

Critical Flow: Raw Text to Structured Metadata

Input: Raw Incident Data

Paste news article or incident report here:

https://cyberscoop.com/grok-undressed-victims-file-class-action-lawsuit-against-xai-elon-musk/

Extract Intelligence (Llama 3)

Output: Structured Intelligence

Extraction Complete


```
{
  "incident_id" : "STNL-2024-001"
  "detected_model" : "Flux.1"
  "method" : "LoRA Fine-tuning"
  "platform_vector" : "Telegram"
  "taxonomy_category" : "NCEI / Targeted Extortion"
  "severity_score" : 8.5
}
```


Verification: Model version confirmed via technical vector analysis.

- Sentinel allows for easy and quick access to structured metadata via Llama 3, reducing latency to <2 seconds.
- Need for an unfiltered, technical LLM layer to bypass commercial safety blocks for forensic use

Risks, Safeguards & Path to CP3

RISK 1: LLM Hallucination in Incident Extraction

Automated metadata extraction may produce incorrect model identification or method classification.

Safeguard:

- → Human-in-the-loop verification
- → Cross-checking with multiple sources
- → Confidence scoring for extracted metadata

RISK 2: Dataset Bias & Incomplete Incidents

News reports and public data may not capture the full technical details of NCEI incidents.

Safeguard:

- → Expand data sources beyond news (forensics reports, research datasets)
- → Authentication and usage logging

RISK 3: Misuse of the Platform

Because Sentinel analyzes the technical mechanics of NCEI attacks, malicious actors could potentially misuse the platform to learn how such attacks are created or optimized.

Safeguard:

- → Restrict access to verified researchers
- → Cross-checking with multiple sources
- → Confidence scoring for extracted metadata

Path to CP3

- → Build a working end-to-end Sentinel prototype
- → Convert a real NCEI incident report → structured metadata using an LLM extraction pipeline
- → Store extracted data in a database and display it in the Sentinel dashboard
- → Provide GitHub repository with setup instructions, prompts, architecture docs, and minimal tests

THANK YOU

Any questions?